How Do Hindi Speakers Search Their Mental Lexicons?

Evidence From Verbal Fluency and Spatial Arrangement Tasks

BRSM Hindi Fluency Project

***Abstract*—This paper studies how Hindi speakers retrieve words from memory. We analyze two tasks: Verbal Fluency Task (VFT) and Spatial Arrangement Method (SpAM). The dataset has 35 participants and 1040 responses. For primary statistical tests, we use only Hindi/Hinglish responses ($n = 723$), and we keep English responses only for comparison context. We test domain-wise differences with Kruskal–Wallis tests, association patterns with Spearman correlation, and VFT sequence dynamics using within-cluster versus cluster-switch transitions. For cluster comparison, we use Adjusted Rand Index (ARI) and Normalized Mutual Information (NMI) between participant SpAM clusters and embedding-based semantic and phonetic clusters. Domain-level tests are mostly not statistically significant, but descriptive patterns are stable across tasks. We also build an equal-weight composite fluency score that combines VFT and SpAM features. The findings suggest that lexical search is structured, but the strength of structure is heterogeneous across participants and domains.**

I. Introduction

The main research question is: *How do Hindi speakers search their mental lexicons for information?*

Verbal fluency performance is often used to understand memory search and lexical organization. Recent work also shows that phonological information can support semantic retrieval [1]. At a broader lexical level, semantic similarity and word-form similarity can co-occur across languages [2]. For spatial data collection of semantic structure, SpAM provides an efficient way to capture perceived similarity in two-dimensional layouts [3].

In this study, we combine VFT and SpAM. The motivation is simple: VFT gives retrieval dynamics (how many words, how fast), while SpAM gives organization dynamics (how words are grouped in space). Together, these two tasks can provide a richer view of lexical search.

A. Research Questions

To avoid repetition, the focused research questions are presented once in Table I.

B. Hypotheses

We explicitly define null and alternative hypotheses for confirmatory analyses.

- **H0$_1$:** Domain distributions of VFT word count are equal. **H1$_1$:** At least one domain differs.
- **H0$_2$:** Domain distributions of VFT mean retrieval time are equal. **H1$_2$:** At least one domain differs.
- **H0$_3$:** Domain distributions of SpAM compactness (mean nearest-neighbor distance) are equal. **H1$_3$:** At least one domain differs.
- **H0$_4$:** Positive ARI alignment occurs with probability 0.5 (no directional tendency). **H1$_4$:** Positive ARI alignment occurs with probability > 0.5.
- **H0$_5$:** Domain distributions of VFT cluster-switch rate are equal. **H1$_5$:** At least one domain differs.
- **H0$_6$:** There is no monotonic association between VFT total output and Hindi confidence. **H1$_6$:** A monotonic association exists.

II. Data and Task Design

A. Participants and Responses

The processed dataset contains 35 sessions (one per participant) and 1040 rows. The domain labels are: *animals*, *body-parts*, *colours*, and *foods*. Language labels in the processed data are Hindi/Hinglish (723 rows) and English (317 rows).

Primary statistical analyses use only Hindi/Hinglish rows. English rows are not used for primary hypothesis testing.

B. Two Tasks

VFT: Participants produced as many words as possible within one minute for each category.

SpAM: Participants arranged generated words in 2D space using normalized x, y coordinates based on perceived similarity.

III. Methods

A. Feature Construction

For each participant-domain unit, we computed:

- **VFT features:** word count, lexical diversity, mean retrieval time.
- **VFT transition features:** within-cluster transitions and cluster-switch transitions over consecutive produced words.
- **SpAM features:** mean nearest-neighbor distance in x, y space, participant cluster labels from hierarchical clustering, and agreement with semantic/phonetic cluster labels.

B. Semantic and Phonetic Embedding Creation

This part is important, so we describe it step by step.

Step 1: Text normalization. Each word was converted to Unicode NFKC form, lowercased, and stripped of extra spaces.

TABLE I
RESEARCH QUESTIONS AND SIMPLE MEANING

RQ	Formal research question	Simple meaning
RQ1	Do VFT productivity and retrieval speed differ across semantic domains?	Are some domains easier or faster for word retrieval?
RQ2	Which participant-level variables are associated with VFT outcomes?	Which personal factors are related to better or worse VFT performance?
RQ3	Does SpAM spatial compactness differ across domains?	Do some domains form tighter semantic maps in SpAM?
RQ4	How strongly do participant SpAM clusters align with semantic and phonetic embedding-based clusters?	Do human clusters match model-based meaning/sound clusters?
RQ5	Does a combined VFT+SpAM fluency score improve inference compared to VFT-only scoring?	Is combined scoring more informative than VFT-only scoring?

Step 2: Semantic embedding with a multilingual transformer. We generated dense sentence-level embeddings for normalized tokens using the Hugging Face Sentence-Transformers model `paraphrase-multilingual-MiniLM-L12-v2`. This model provides a shared multilingual semantic space suitable for short Hindi/Hinglish lexical items [4].

Step 3: Phonetic key generation. For each token, we generated a phonetic key using a deterministic Devanagari-to-Latin mapping. Then we removed vowels and collapsed repeated consonant symbols to reduce spelling noise and keep broad sound structure.

Step 4: Phonetic embedding with the same transformer encoder. We encoded phonetic keys using the same multilingual transformer. This keeps the representation family consistent across semantic and phonetic channels while changing input form (surface word vs phonetic key).

Step 5: Cluster generation and alignment. For each participant-domain unit:

- SpAM clusters were built from (x, y) coordinates using Ward hierarchical clustering.
- The number of clusters k was selected by silhouette criterion over candidate values.
- Semantic and phonetic embeddings were clustered using K-means with the same k.

Step 6: Similarity metrics. We compared cluster partitions using ARI [8] and NMI (normalized information overlap; [10]). Cluster-quality selection used silhouette-guided k selection [9].

C. Statistical Tests

To stay aligned with BRSM methods and avoid assumption mismatch, test selection followed a normality-first rule:

- For each domain-comparison endpoint, we first ran Shapiro–Wilk within each domain [5].
- If all groups were approximately normal ($p > 0.05$), we used one-way ANOVA.
- If at least one group was non-normal, we used Kruskal–Wallis [6].
- For participant-level associations, we used Spearman rank correlation [7] because several involved variables were non-normal and relationships were treated as monotonic rather than strictly linear.
- For directional tendency of positive ARI values, we used a binomial sign test.

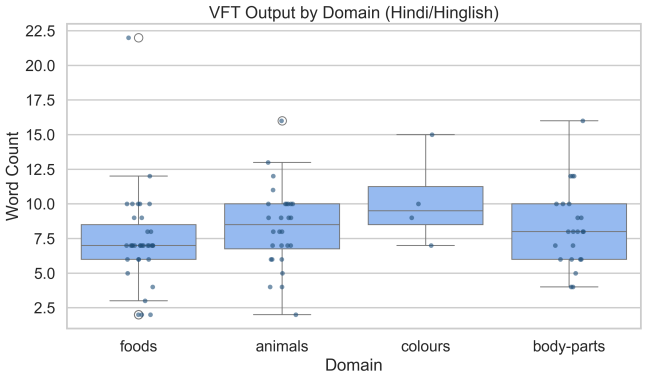

Fig. 1. VFT word count by domain (Hindi/Hinglish subset).

D. Composite Fluency Score

We define a participant-domain score using standardized components:

$$S_d = \frac{1}{6}\left(z_{\text{word}} + z_{\text{diversity}} + z_{\text{speed}} + z_{\text{spatial}} + z_{\text{semAlign}} + z_{\text{phoAlign}}\right)$$

where all domains have equal weight. The participant score is:

$$S_{\text{participant}} = \frac{1}{D}\sum_{d=1}^{D} S_d$$

with D equal to number of available domains for that participant.

IV. RESULTS

A. Descriptive Patterns

Table III reports domain-level descriptive values for Hindi/Hinglish rows.

Interpretation. The descriptive view shows domain variability in output size, with colours having higher mean output in this sample. But this descriptive pattern still needs inferential testing.

TABLE II
NORMALITY-FIRST TEST DECISION FRAMEWORK USED IN THIS STUDY

Analysis target	Decision rule	Test used in final analysis
Domain-wise VFT word count	At least one domain failed Shapiro–Wilk	Kruskal–Wallis
Domain-wise VFT mean RT	At least one domain failed Shapiro–Wilk	Kruskal–Wallis
Domain-wise VFT cluster-switch rate	At least one domain failed Shapiro–Wilk	Kruskal–Wallis
Domain-wise SpAM compactness (mean NN distance)	At least one domain failed Shapiro–Wilk	Kruskal–Wallis
Participant-level outcome associations	Non-normality and monotonic relation target	Spearman correlation
ARI directional tendency	Sign direction over paired outcomes	Binomial sign test

TABLE III
DOMAIN-LEVEL SUMMARY (HINDI/HINGLISH SUBSET)

Domain	Sess.-dom N	Mean words	Mean RT (ms)	Mean lex. div.
colours	4	10.25	5392.52	0.914
animals	28	8.321	7196.57	0.996
body-parts	23	8.304	7384.59	0.995
foods	35	7.371	6978.33	1.000

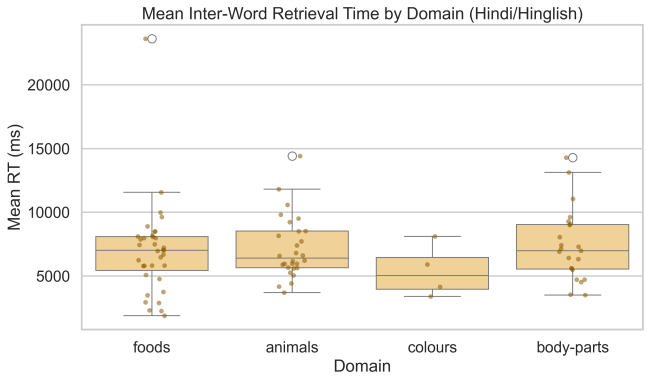

Fig. 2. Mean retrieval time by domain (Hindi/Hinglish subset).

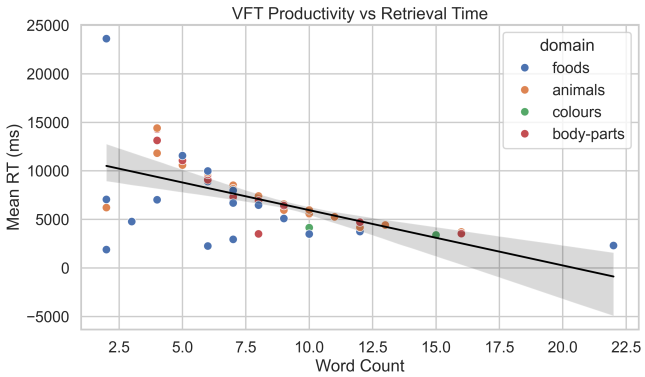

Fig. 3. Relation between VFT output size and retrieval speed.

B. VFT Domain Tests

Question addressed (RQ1): Do VFT productivity and retrieval speed differ across domains?

Based on the normality-first rule, Shapiro–Wilk indicated at least one non-normal domain distribution for both endpoints, so Kruskal–Wallis was applied.

- Word count by domain: $H = 5.295$, $p = 0.1515$ (not significant).
- Mean RT by domain: $H = 1.919$, $p = 0.5894$ (not significant).

Inference. For RQ1, we do not reject the null hypothesis at $\alpha = 0.05$. Domain differences are present descriptively, but inferential evidence is not strong in this sample.

C. VFT Transition Dynamics (Within-Cluster vs Cluster-Switch)

This subsection addresses the missing process-level part of VFT analysis by examining how participants moved through clusters over time.

For each participant-domain sequence, words were ordered by production position. Using participant SpAM-derived clusters in that domain, each adjacent word pair (w_t, w_{t+1}) was labeled as:

- **within-cluster transition** if both words had the same cluster label,
- **cluster-switch transition** if labels differed.

Domain differences in switch-rate were tested with Kruskal–Wallis (normality-first rule): $H = 1.909$, $p = 0.5916$.

Exploratory session-level associations were also weak and non-significant:

- total words vs switch-rate: $\rho = -0.043$, $p = 0.8147$;

TABLE IV
VFT TRANSITION SUMMARY BY DOMAIN

Domain	N	Mean words	Mean within	Mean switch	Within rate	Switch rate
colours	4	10.250	4.500	4.750	0.456	0.544
animals	27	8.556	4.037	3.519	0.532	0.468
foods	31	8.032	3.871	3.161	0.537	0.463
body-parts	23	8.304	4.391	2.913	0.591	0.409

- Hindi confidence vs switch-rate: $\rho = 0.180$, $p = 0.3230$.

Inference. Within-cluster and cluster-switch transitions are now explicitly quantified, but in this sample they do not show strong domain effects or strong association with confidence. Thus, transition behavior should be interpreted as descriptive support for strategy heterogeneity, not as a confirmed group-level effect.

D. Associations With Background Variables

Question addressed (RQ2): Which participant-level variables are associated with VFT outcomes?

Spearman correlation was selected because normality assumptions were not consistently satisfied and the target relation was monotonic.

At session level, significant Spearman results include:

- Total words vs Hindi confidence: $\rho = -0.395$, $p = 0.019$.

Interpretation. This indicates a moderate negative monotonic relationship: participants with higher total VFT output tended to report lower Hindi confidence in this sample.

The direction is counter-intuitive and should not be over-interpreted. A plausible explanation is measurement mismatch: confidence is self-reported and can reflect perceived ability rather than objective lexical performance. More fluent participants may under-rate themselves, while less fluent participants may over-rate themselves. Another possibility is strategy variation: some lower-confidence participants may produce more words through risk-taking, whereas highly confident participants may answer more cautiously. Finally, the modest sample size ($n = 35$) increases sensitivity to outliers and sampling variation.

Inference. For RQ2, there is evidence of a statistically significant participant-level association, but the negative direction is not theoretically straightforward and is likely influenced by self-report bias, strategy differences, and sample-size limitations.

E. SpAM Domain Test

Question addressed (RQ3): Does SpAM spatial compactness differ across domains?

Shapiro–Wilk diagnostics indicated at least one non-normal domain distribution for compactness, so Kruskal–Wallis was used.

Mean nearest-neighbor distance by domain gave $H = 3.038$, $p = 0.3857$ (not significant).

Inference. For RQ3, domain-wise compactness differences are not statistically significant, so we treat them as descriptive only.

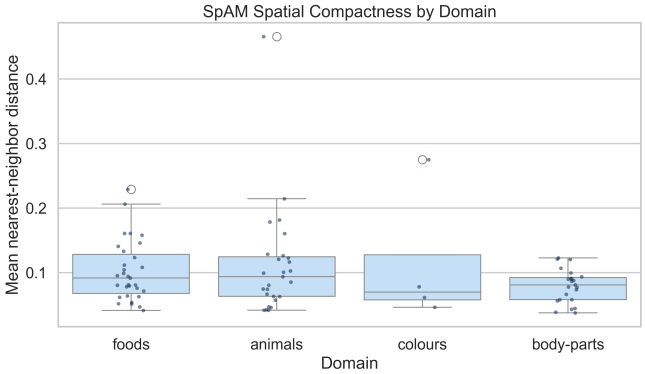

Fig. 4. SpAM compactness (mean nearest-neighbor distance) by domain.

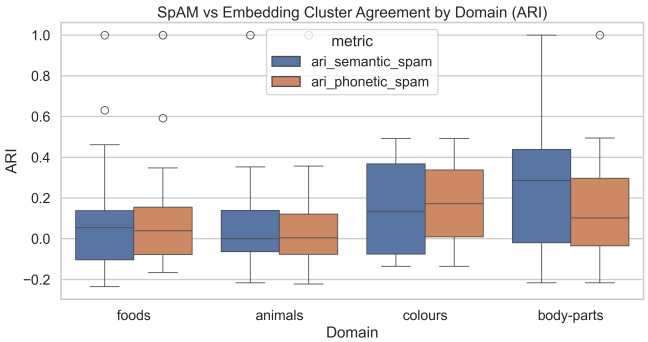

Fig. 5. Domain-wise ARI alignment between participant SpAM clusters and embedding-based clusters.

F. Cluster Similarity Metrics

Question addressed (RQ4): How strongly do participant SpAM clusters align with semantic and phonetic embedding-based clusters?

Table V shows domain-wise mean agreement values.

Global means were: ARI(sem,SpAM)=0.154, ARI(pho,SpAM)=0.111, NMI(sem,SpAM)=0.405, NMI(pho,SpAM)=0.364.

Sign tests for positive ARI tendency were significant:

- semantic-SpAM ARI: 48/78 positive, $p = 0.0268$.
- phonetic-SpAM ARI: 48/79 positive, $p = 0.0356$.

Inference. For RQ4, sign tests and NMI both support non-trivial positive alignment with semantic and phonetic representations, with stronger average overlap in the semantic channel.

TABLE V
DOMAIN-WISE MEAN CLUSTER SIMILARITY METRICS

Domain	ARI(sem,SpAM)	ARI(pho,SpAM)	NMI(sem,SpAM)	NMI(pho,SpAM)	Mean NN distance
foods	0.086	0.106	0.364	0.370	0.102
body-parts	0.316	0.162	0.490	0.340	0.079
colours	0.157	0.175	0.301	0.319	0.115
animals	0.095	0.064	0.394	0.384	0.111

Fig. 6. Average ARI and NMI values for semantic and phonetic cluster comparisons.

G. Composite Fluency Score

Question addressed (RQ5): Does combined VFT+SpAM scoring improve over VFT-only scoring?

The integrated score combines VFT and SpAM features with equal domain weights.

For comparison with self-reported Hindi confidence:

- VFT-only score: $\rho = -0.449$, $p = 0.0069$.
- Integrated score: $\rho = -0.317$, $p = 0.0633$.

Inference. For RQ5, integrated scoring does not show stronger association with confidence than VFT-only scoring in this dataset and is not significant at $\alpha = 0.05$. However, it still adds structure-sensitive information from SpAM and may be useful for broader fluency profiling.

H. Question-wise Summary

Table VI provides direct mapping from each question to statistical outcome and inference.

V. DISCUSSION

The results suggest a structured lexical search process, but with mixed statistical strength across analyses.

First, domain-wise inferential tests are mostly non-significant. This means we should avoid strong claims about stable domain differences for this sample size.

Second, transition-level VFT analysis shows measurable within-cluster and switch behavior, but no strong domain-level switch-rate effect. This is still informative: lexical search strategy appears heterogeneous across individuals, even when group-level contrasts are weak.

Third, the significant negative association between output size and self-reported confidence should be interpreted cautiously. It likely reflects a gap between perceived and actual performance, together with strategy differences and sampling variability.

Fourth, SpAM and embedding clusters show moderate information overlap (NMI), and sign tests indicate positive ARI tendency in both semantic and phonetic channels. This indicates systematic structure linkage, but also leaves room for participant-specific strategies not captured by one global clustering rule.

Finally, adding SpAM features gives a broader fluency profile, even though it did not increase correlation with confidence in this dataset. In simple words, integrated scoring gives richer structure, but not necessarily stronger external association for every target variable.

VI. CONCLUSION

This study combines VFT and SpAM to answer how Hindi speakers search their mental lexicons. The evidence supports a structured search process with participant-level heterogeneity. We find descriptive domain patterns, moderate semantic/phonetic-spatial overlap, and meaningful but mixed cross-task coupling. So, lexical search appears organized, but the organization is not uniform across all participants and all domains.

LIMITATIONS

- Sample size is moderate (35 participants), which limits power for subgroup effects.
- Phonetic representation is simplified, and richer phonological modeling can improve precision.
- External validation targets are limited in the current dataset.

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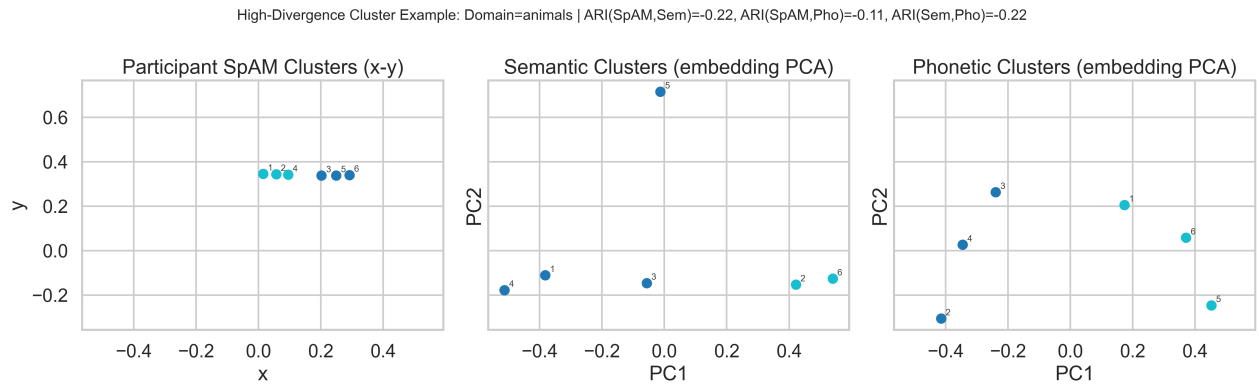

Fig. 7. Example participant-domain case: SpAM clusters versus semantic and phonetic cluster assignments.

TABLE VI
RESEARCH QUESTION, RESULT, AND INFERENCE SUMMARY

RQ	Question	Main result	Inference
RQ1	VFT output/speed by domain	Word count: $p = 0.1515$; RT: $p = 0.5894$	No significant domain effect
RQ2	Participant factors vs VFT outcomes	$\rho_{\text{words,conf}} = -0.395$ ($p = 0.019$)	Significant association; direction needs caution
RQ3	SpAM compactness by domain	Kruskal–Wallis $p = 0.3857$	No significant domain effect
RQ4	SpAM alignment with semantic/phonetic clusters	ARI sign tests significant; NMI $\approx$ 0.36–0.41	Positive directional evidence; moderate overlap
RQ5	Integrated score vs VFT-only score	$\lvert\rho_{\text{integrated,conf}}\rvert < \lvert\rho_{\text{VFT,conf}}\rvert$, $p_{\text{integrated}} > 0.05$	Structurally richer, not stronger for confidence proxy
S1	VFT transition dynamics (supplementary)	Switch-rate Kruskal–Wallis: $H = 1.909$, $p = 0.5916$	Measurable transitions; effects remain descriptive

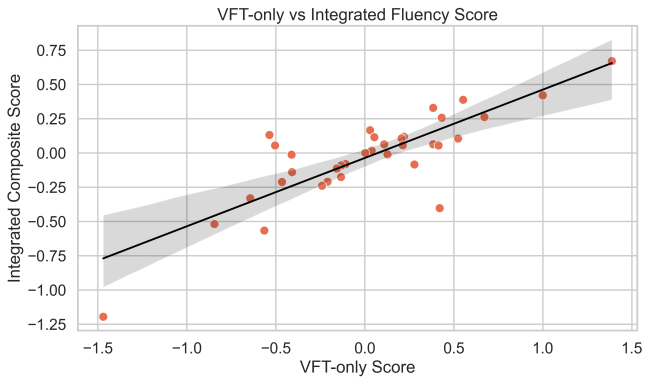

Fig. 8. VFT-only score versus integrated (VFT+SpAM) composite score.

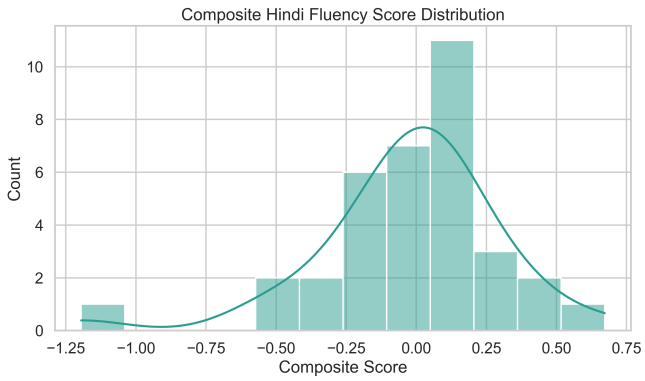

Fig. 9. Distribution of integrated composite Hindi fluency score.

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